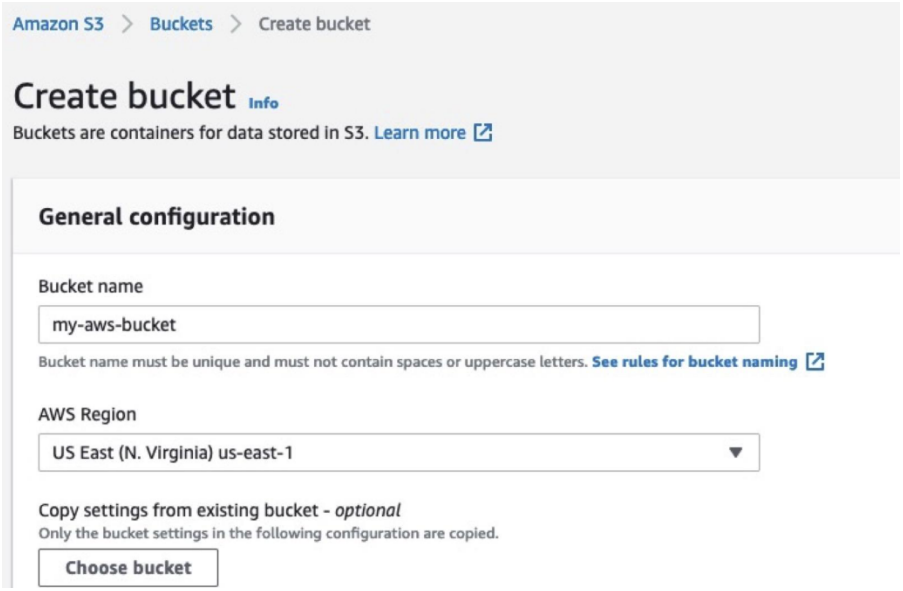
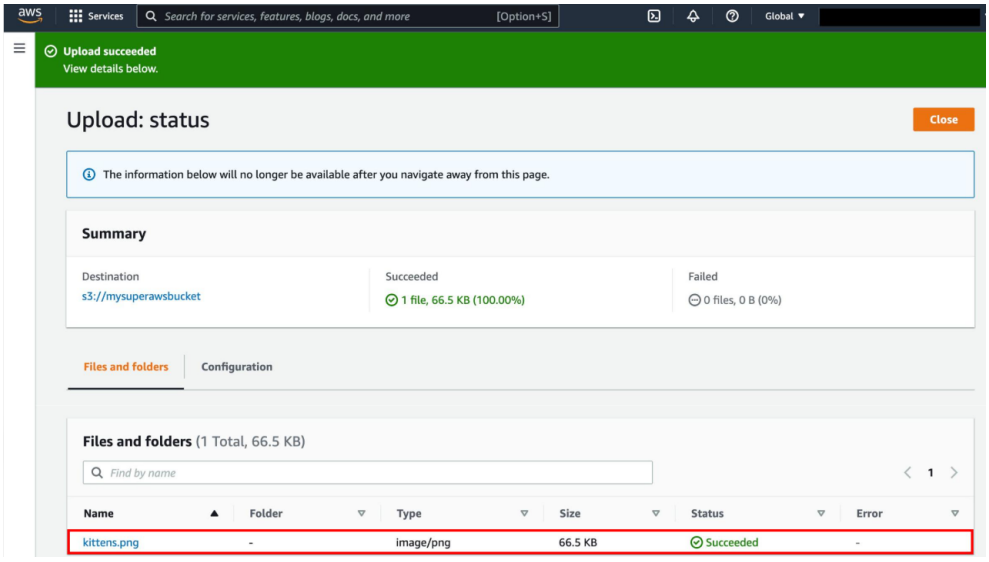
1.Create a new S3 bucket on AWS named after your favorite food delivery app (e.g., swiggy-files), and upload any one image or PDF file from your computer to this bucket.

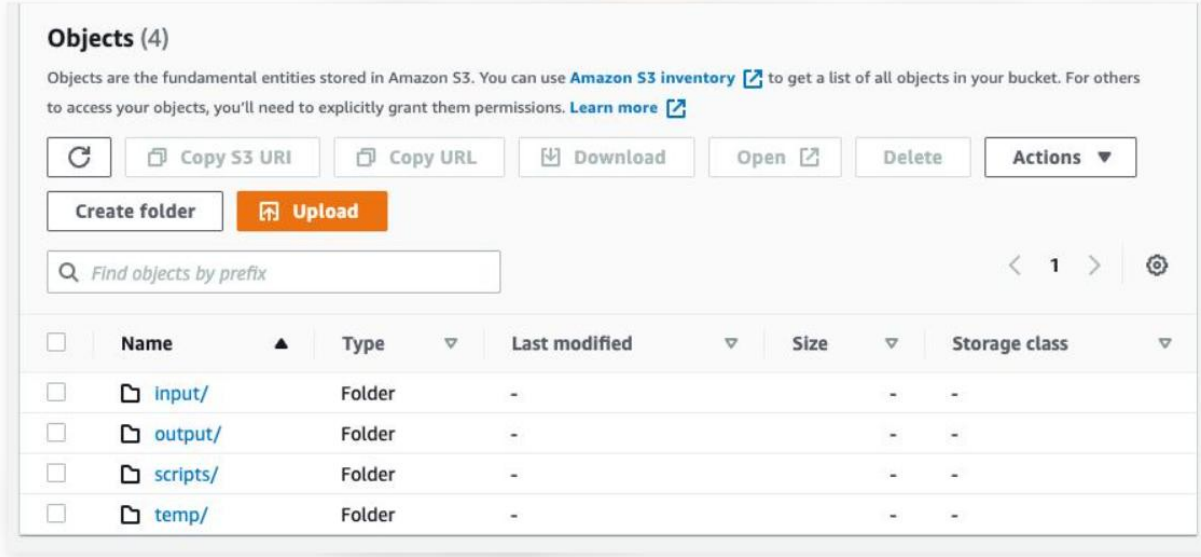
Ans-





2.Set up a folder structure inside your S3 bucket that mimics a music streaming app like Spotify: create folders named 'albums', 'artists', and 'playlists', and upload at least one sample file into each folder.

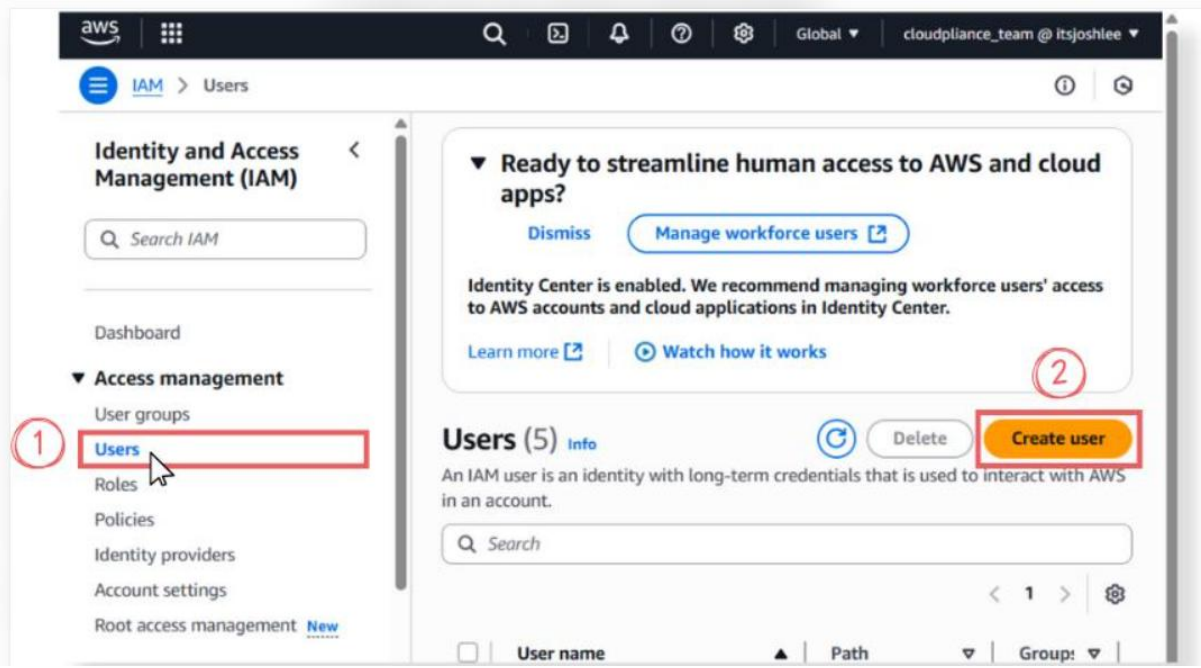
Ans-

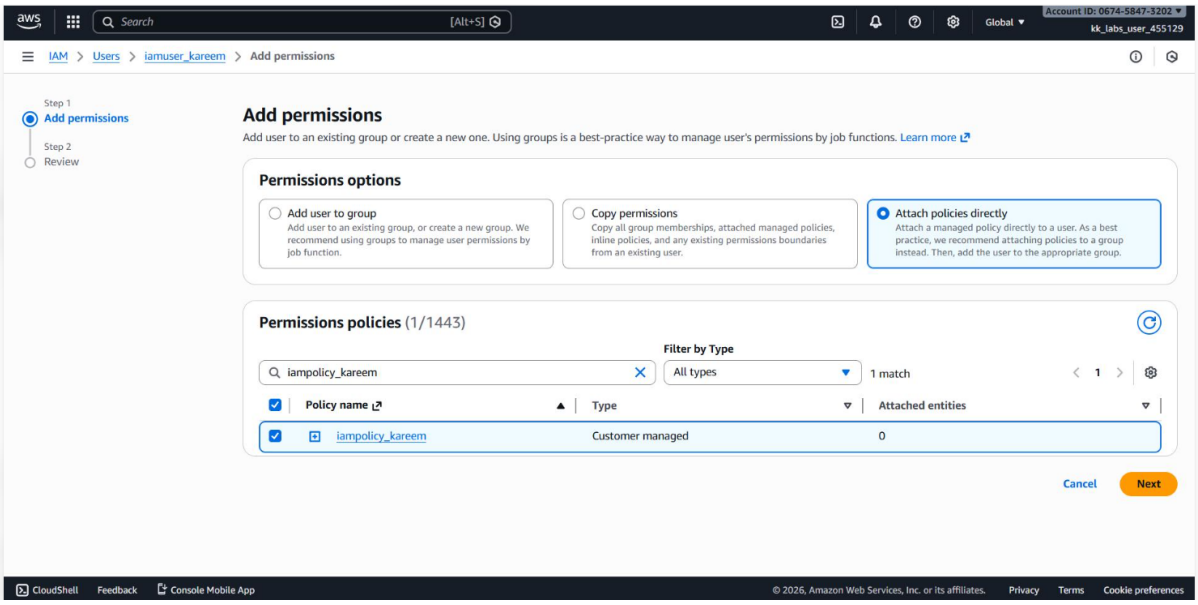


**3.Configure an IAM user named 'zomatoUploader' and grant it permission to only upload and list files in your S3 bucket, but not delete or download files.

Hint: Use the AWS IAM console to attach a custom policy with s3:PutObject and s3:ListBucket permissions only.**

Ans-





4.Test your IAM setup by logging in as 'zomatoUploader' and trying to upload a file and then delete a file from your S3 bucket; note which actions succeed and which are denied.

Ans-Step 1: Get IAM sign-in details

1. Open IAM.
2. Select zomatoUploader.
3. Find the AWS account sign-in URL.
4. Copy the URL.
5. Open an incognito/private browser window.
6. Log in using the IAM username and password.

Step 2: Test upload permission

1. Open the S3 bucket zomato-files-2026.
2. Try uploading a small image or text file.
3. Click Upload.
4. Check whether the upload succeeds.

Expected result:

Allowed

Step 3: Test list permission

1. Open the bucket.
2. View its existing objects.

3. Check whether the files and folders are displayed.

Expected result:

Allowed

Step 4: Test delete permission

1. Select an uploaded test file.
2. Click Delete.
3. Confirm the delete operation.
4. Observe the error message.

Expected result:

Access Denied

The file should remain in the bucket.

Step 5: Test download permission

1. Select an existing file.
2. Try downloading it.
3. Observe the result.

Expected result:

Access Denied

5.Write a short note explaining how IAM roles and policies in AWS help protect user data in cloud apps like Paytm or PhonePe, and give one real-world example of a security risk if IAM is not configured properly.

Ans-

IAM (Identity and Access Management) helps protect cloud applications by controlling who can access resources and what actions they can perform.

Example: Paytm or PhonePe

- IAM ensures that only authorized employees and applications can access customer information.
- It restricts access to sensitive data such as transaction records and payment information.
- IAM policies provide permissions according to job responsibilities.
- IAM roles allow applications and AWS services to access required resources without sharing permanent user credentials.

Real-world security risk:

Suppose a company accidentally gives an employee administrator permissions to an S3 bucket containing customer transaction records. The employee or an attacker using the compromised account could download, delete or expose sensitive customer information.

Prevention:

- Follow the Principle of Least Privilege.
- Use MFA (Multi-Factor Authentication).
- Regularly review IAM permissions.
- Avoid giving unnecessary administrator access.
- Enable logging and monitoring using AWS CloudTrail.

Conclusion:

IAM roles and policies help protect cloud applications by ensuring that users and services access only the resources they need. Proper IAM configuration reduces unauthorized access, data leaks and security risks.